

Ferna PDF Reader

Installation & User Manual

Version 1.4.0 · September 2026

For Brave, Google Chrome, Microsoft Edge and Mozilla Firefox on Windows

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Ferna PDF Reader is a browser extension that opens PDFs with every page fitted to the window width, and lets you rotate pages, annotate, write with a stylus, save straight over the original file, password-protect PDFs, preview cross-references, generate bookmarks, and summarise, question and quiz you on documents with an on-device AI — in English and Sinhala.

1. Requirements

You need a Chromium-based browser and, for direct saving of PDFs on your disk, Windows 10 or 11.

Item	Requirement
Browser	Brave, Google Chrome or Microsoft Edge, version 120 or later (desktop)
Operating system	Windows 10/11 for the direct-save helper; the reader itself also runs on macOS and Linux
PowerShell	Windows PowerShell 5.1 — already part of Windows 10/11
Rights	Normal user account; no administrator rights needed
Graphics (AI features only)	A graphics card with WebGPU and 1–4 GB free graphics memory — or a local AI server (see section 11)
Package	`ferna-pdf-reader-1.3.0.zip`

The zip contains one folder, `ferna-pdf-reader`, with these parts:

Folder / file	Purpose
`manifest.json`, `background.js`	The extension itself
`viewer/`	The PDF reader
`lib/`	pdf.js (display), pdf-lib (saving, encryption) and WebLLM (on-device AI)
`save-helper/`	`Install-Save-Helper.bat`, `Uninstall-Save-Helper.bat` and the helper script
`README.md`	Short technical notes

2. Installing the extension

The extension is installed as an "unpacked" extension from a folder on your computer. The steps are the same in every Chromium browser; only the address of the extensions page differs.

Browser	Extensions page
Brave	`brave://extensions`
Google Chrome	`chrome://extensions`
Microsoft Edge	`edge://extensions`

1. **Unzip the package** to a permanent place, for example `Documents\Extensions\ferna-pdf-reader`. The browser loads the extension from this folder every time it starts, so do not move or delete it later.
2. **Open the extensions page** by typing its address (table above) in the address bar.
3. **Switch on Developer mode** with the switch at the top-right (Edge: bottom-left).
4. **Click Load unpacked** and choose the `ferna-pdf-reader` folder — the one that contains `manifest.json`.
5. **Allow local files:** on the Ferna PDF Reader card click **Details** and switch on **Allow access to file URLs**. Without this, PDFs on your disk still open in the browser's own viewer.
6. **Pin the icon:** click the puzzle-piece icon beside the address bar, then the pin next to Ferna PDF Reader.
7. **Brave only:** open `brave://settings/content/pdfDocuments` and choose **Open PDFs in Brave**, so PDF links open instead of downloading.

> **Check it works:** open any online PDF, for example `https://arxiv.org/pdf/1706.03762`. It should open in Ferna, with the blue **Annotate** button at the top-right.

The browser may show a note about extensions in developer mode when it starts. This is normal for extensions loaded from a folder; close it.

Firefox

Ferna PDF Reader is also available for Firefox 140 or later (package `ferna-pdf-reader-1.4.0-firefox.zip`).

1. **From Firefox Add-ons** (once published): open the Ferna PDF Reader page on addons.mozilla.org and click **Add to Firefox**. Firefox lists the permissions; click **Add**.
2. **For testing before publication:** open `about:debugging#/runtime/this-firefox`, click **Load Temporary Add-on…** and choose `manifest.json` in the unzipped Firefox package. A temporary add-on is removed when Firefox closes.
3. **Pin the button:** click the puzzle-piece (Extensions) button and choose **Pin to Toolbar** for Ferna PDF Reader.

PDF links you open in Firefox now open in Ferna instead of Firefox's own viewer.

How Firefox differs:

Topic	In Firefox
PDFs on your disk	Firefox never lets an extension read files on your disk. Install the save helper (section 3, from the separate `Ferna-PDF-Save-Helper-1.4.0.zip`): PDFs you open from your disk then open in Ferna, **Open** shows the Windows Open window, and **Save** overwrites the file. Without the helper, use **Open** or drag a PDF into the window; Save then writes a copy to Downloads
Save As	The helper's Windows Save As window, or (without the helper) Firefox's download Save As window

Topic	In Firefox
Printing	Ferna prints high-resolution images of the pages
AI	On-device models need WebGPU (Firefox 141 or later on Windows). Otherwise use a local AI server; for Ollama set `OLLAMA_ORIGINS=moz-extension://*`. Chrome's built-in AI is not available

3. Installing the direct-save helper (Windows, once)

The helper lets **Save** write straight over a PDF you opened from your disk, with no File Explorer window. Browsers never allow an extension to write files on its own, so this small helper does the writing for Ferna. Without it, Ferna still works, but saving a disk PDF needs **Save As**.

1. Open the extension folder `ferna-pdf-reader`, then the `save-helper` folder. (Firefox: unzip `Ferna-PDF-Save-Helper-1.4.0.zip` anywhere and open that folder.)
2. Double-click **Install-Save-Helper.bat**.
3. If Windows SmartScreen says "Windows protected your PC", click **More info**, then **Run anyway**.
4. A window reports **Ferna PDF save helper installed**. Press any key to close it.
5. Close every browser window and start the browser again.
6. In Ferna, open **⋮ (More) → Settings**. The row **Direct-save helper** should read **Installed** ✓.

What the installer does:

- Copies the helper to `%LOCALAPPDATA%\FernaPDF\save-helper` (your own user folder).
- Registers it for Chrome, Brave, Edge, Chromium and Firefox under your user account only (`HKEY_CURRENT_USER`).
- Changes nothing else on the computer.

What the helper will and will not do:

- It answers only the Ferna PDF Reader extension; websites and other extensions cannot use it.
- It only overwrites files that already exist and end in `.pdf`.
- It writes to a temporary file first and swaps it in at the end, so a failed save never damages the original.

To remove it: double-click **Uninstall-Save-Helper.bat** in the same folder.

4. Updating and uninstalling

Updating from 1.2 to a later version: unzip the new version over the same folder (replace all files), then click the round reload arrow on the Ferna PDF Reader card in the extensions page. The save helper only needs reinstalling if the release notes say so.

Updating from 1.0 or 1.1 to 1.2: version 1.2 has a fixed extension ID, which the save helper relies on. Do this once:

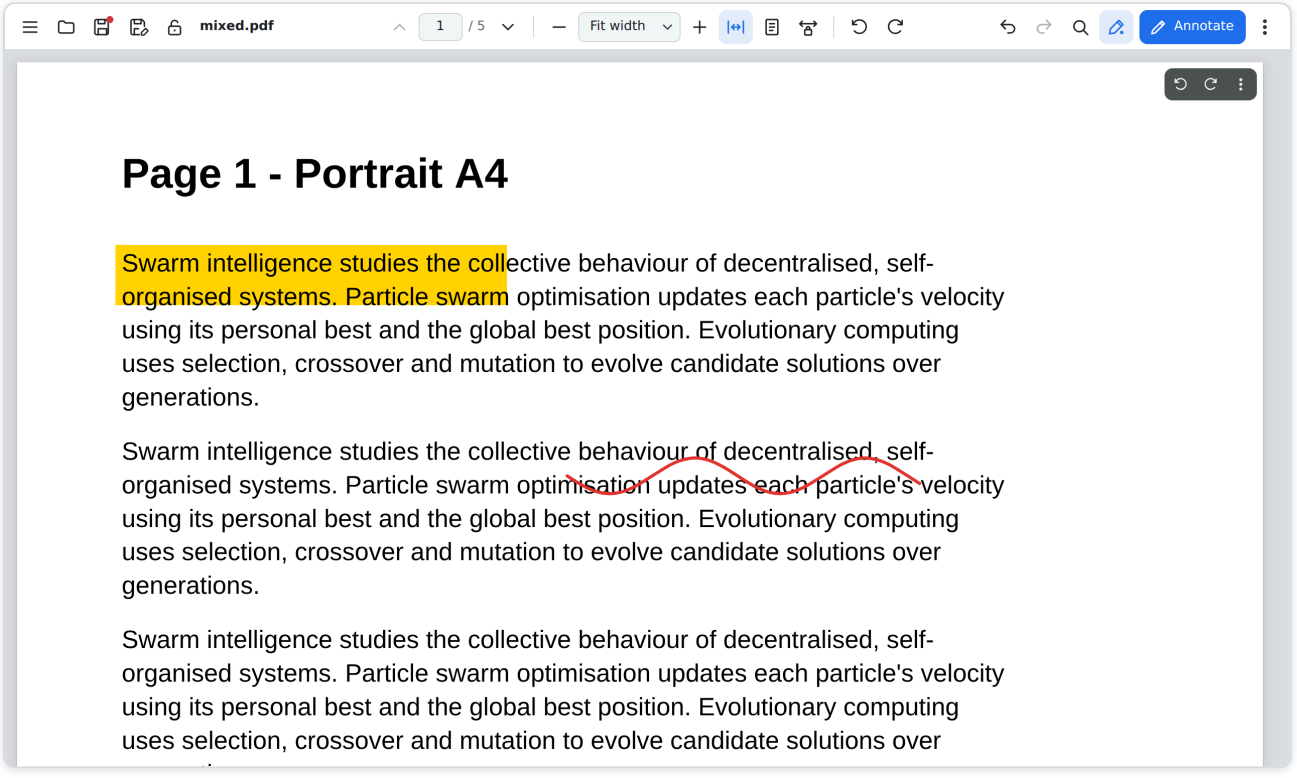
1. On the extensions page, click **Remove** on the old Ferna PDF Reader.
2. **Load unpacked** again and choose the new folder.
3. Switch on **Allow access to file URLs** again.
4. Install the save helper (section 3).

Your Ferna settings return to their defaults once. Annotations are stored inside your PDF files, so none are lost.

Uninstalling: click **Remove** on the extensions page, then run **Uninstall-Save-Helper.bat**. The browser's own PDF viewer takes over again.

5. Opening PDFs and the screen layout

Once installed, PDFs you open in the browser appear in Ferna automatically.



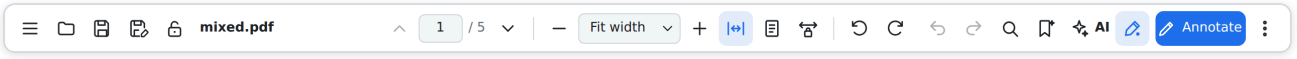
Ferna PDF Reader with an area highlight and pen ink on a page fitted to the window width.

How	What happens
Click a PDF link on a website	Opens in Ferna in the same tab

How	What happens
Double-click a PDF in File Explorer (browser set as PDF app)	Opens in Ferna (needs **Allow access to file URLs**)
Ferna toolbar icon	On a PDF tab: reopens it in Ferna. Elsewhere: opens an empty Ferna tab
Right-click a link → **Open link in Ferna PDF Reader**	Opens that PDF in a new tab
Open button or Ctrl+O	Choose a PDF on your computer
Drag a PDF onto a Ferna tab	Opens it
Right-click the toolbar icon	Turn **Open PDFs automatically in Ferna PDF Reader** on or off, or open a local file

Password-protected PDFs ask for their password first. Ferna remembers the last page and zoom of each document.

The toolbar



The toolbar.

Button	Function
☰ Sidebar	Page thumbnails, outline and annotation list (F4)
Open	Open a PDF from your computer (Ctrl+O)
Save	Overwrite the current file (Ctrl+S); a red dot means unsaved changes
Save As	Save to a new name or place (Ctrl+Shift+S)
Lock	Add, change or remove a password
▲ page box ▼	Previous page, current page number (type and press Enter), next page
− zoom box +	Zoom out, zoom level (type a percentage), zoom in; the arrow opens zoom presets
Fit width / Fit page	Every page to the window width / whole page in view
Horizontal lock	Stop sideways movement when zoomed in (L)
Rotate left / right	Rotate the current page (Shift+R / R)
Undo / Redo	Ctrl+Z / Ctrl+Y
Find	Search the text (Ctrl+F)
Auto-Bookmark	Generate bookmarks for the document (section 10)

Button	Function
✦ AI	Open the AI assistant: Summary, Ask, Quiz (I)
Stylus	Xodo mode on/off: a pen writes without choosing a tool
Annotate	Open the annotation tool panel (A)
⋮ More	Open, Save, Save as, Password, Auto-Bookmark, AI assistant, Download original, Print, Rotate all, Settings, Keyboard shortcuts

On narrow windows some buttons hide; their functions stay in the ⋮ menu and on the keyboard.

6. Reading

Fit width, page by page

The default zoom is **Fit width**: each page is scaled on its own to the width of the window, so portrait and landscape pages in the same document both fill the width. Pages re-fit automatically when you resize the window, open the sidebar or rotate a page.

Zoom

- **Fit width** (W) or **Fit page** (P) buttons.
- **Zoom box:** click it, type a percentage such as `135` and press Enter; the arrow beside it lists 50–400 % and Actual size.
- **Ctrl + mouse wheel**, touchpad pinch or touchscreen pinch zoom around the pointer.
- **Ctrl +** / **Ctrl −** step the zoom; **Ctrl 0** returns to Fit width; **Ctrl 1** is 100 % (real size).

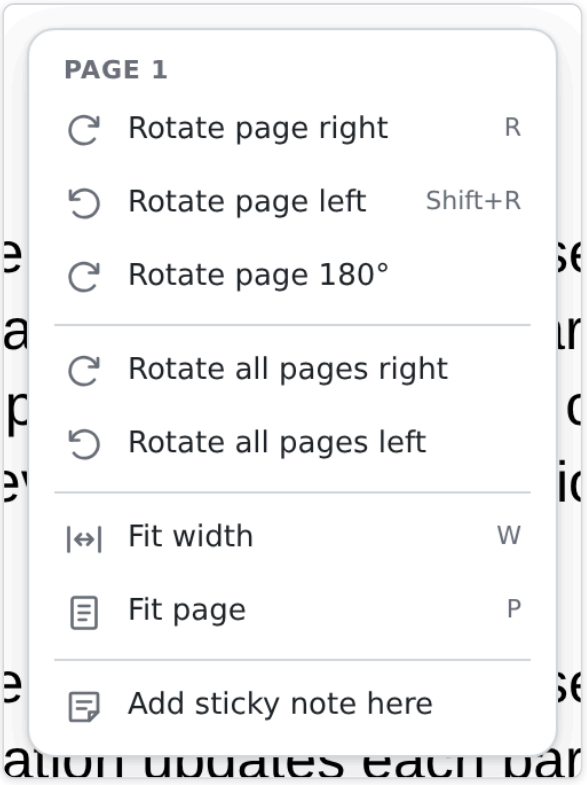
Horizontal lock

When zoomed in, click **Horizontal lock** (or press L) to stop the pages moving sideways. Touchpad swipes, Shift + wheel, touch and the Hand tool then scroll only up and down. First scroll to the part of the page you want, then lock. Press L again to unlock. The setting is remembered.

Moving through the document

- Scroll normally, or use Page Up / Page Down and Home / End.
- ← / → go to the previous / next page when the page fits the width.
- Type a page number in the page box and press Enter.
- Links inside the PDF are clickable. Cross-references (figures, tables, equations, citations) open a preview card instead of jumping away — see section 9.

Rotating pages



The page menu (right-click a page).

- **Rotate left / right** in the toolbar, or Shift+R / R, rotate the current page.
- Hover over a page: small rotate buttons appear at its top-right corner.
- **Right-click a page** (long-press on a touchscreen) for the page menu: rotate right, left or 180°, reset rotation, rotate all pages, Fit width, Fit page and Add sticky note here.
- Rotations can be undone (Ctrl+Z) and are saved into the PDF.

Sidebar

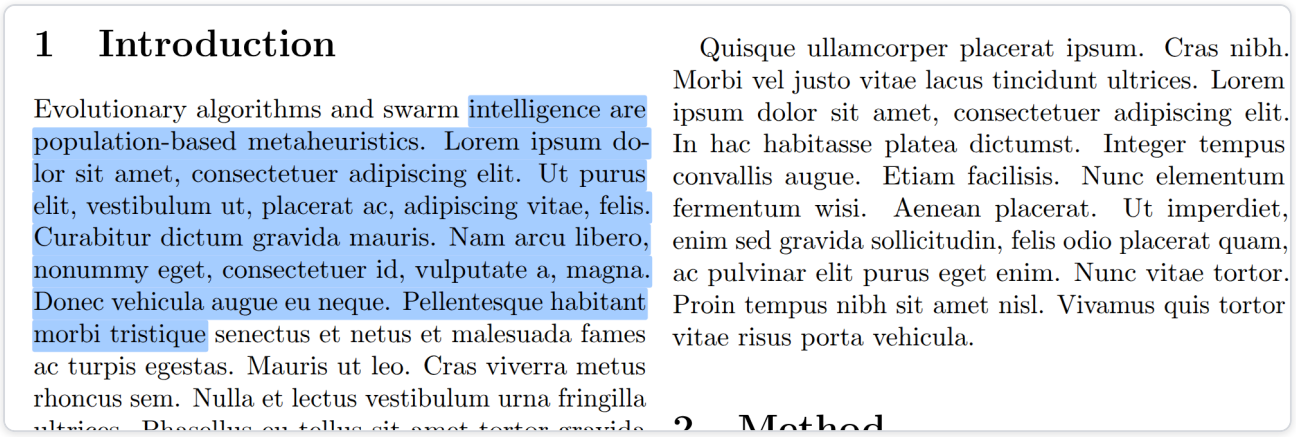
Press F4 or click ☰. **Pages** shows thumbnails (click to jump), **Outline** shows the document's bookmarks, **Annotations** lists every annotation with its page and comment — click one to go to it.

Selecting text

Ferna has its own text selection, built to feel like Xodo:

- **Drag** across text. The selection follows the character nearest to the pointer, even when the pointer is between lines, between words or in the margin, so it never jumps to the end of the page.
- **Whole words:** once the selection leaves the first word it grows word by word, so words are never cut in half — also in LaTeX documents. Hold **Alt** while dragging to select single characters.
- **Double-click** selects a word, **triple-click** a line. **Shift+click** extends the current selection.

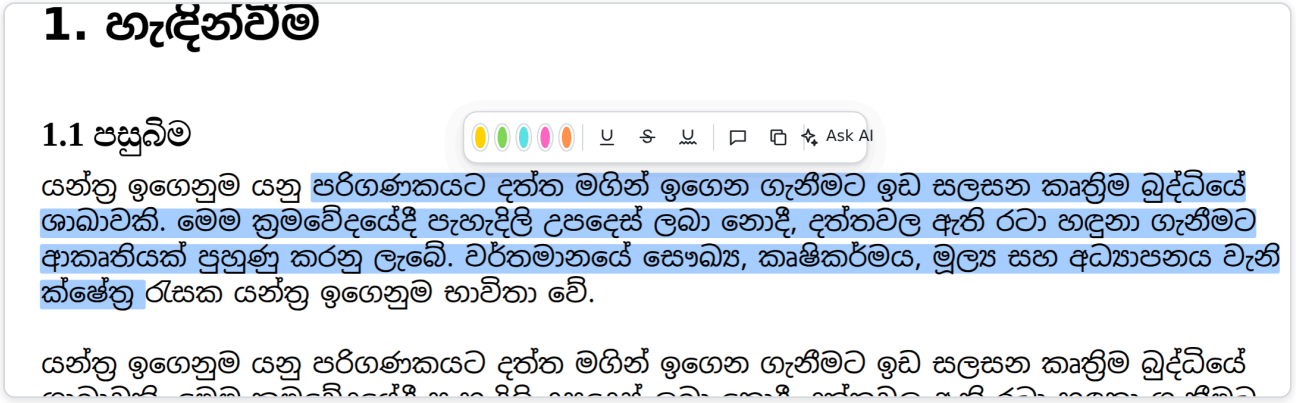
- Drag towards the top or bottom edge of the window to scroll automatically.
- The selection is drawn as one clean band per line. Highlights made from it (quick popup or Highlight tool) have the same neat shape.
- Ferna reads the exact word positions from the rendered page, so the selection lines up with the printed words.



Selecting in a two-column LaTeX paper: whole words, one band per line.

Sinhala text. Many PDFs store Sinhala in a form that other viewers show garbled: vowel signs such as ෙ before their consonant ("ෙක" instead of "කෙ"), missing letters such as the rakaransaya (්‍ර), or the real text hidden in a special PDF field (LibreOffice and Word do this). Ferna repairs all of these, so selecting, copying, searching, the AI and Auto-Bookmark see real Sinhala words. Where a PDF places Sinhala letters one by one, Ferna takes the word spaces from the printed page.

Settings → **Smooth text selection** and **Select whole words when dragging** switch these features off.



Sinhala text selected in a PDF made by a word processor, with the quick popup.

Find

Press Ctrl+F, type a word or phrase (English or Sinhala). Matches are highlighted in orange; Enter / Shift+Enter move to the next / previous match; **Match case** limits it to exact capitals. Esc closes the find bar.

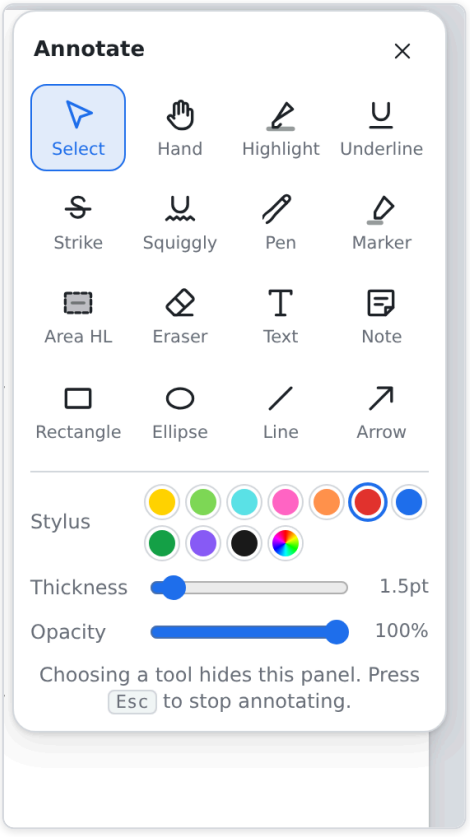
7. Annotating

The tool panel hides itself

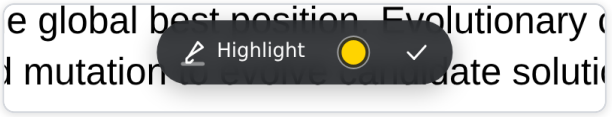
Click **Annotate** (or press A) to open the tool panel. As soon as you pick a tool, the panel disappears so the page is clear, and a small dark bar at the bottom of the screen shows the active tool and its colour:

- click the tool name on the bar to bring the panel back;
- click the colour dot to change colour;
- click ✓, or press Esc, to stop annotating and return to Select.

The lower part of the panel sets the tool's **colour** (ten colours plus a custom colour), **thickness**, **opacity**, **font size** (Text) and **fill** (Rectangle, Ellipse). Each tool remembers its own settings.



The annotation tool panel.



The bar that replaces the panel while a tool is active.

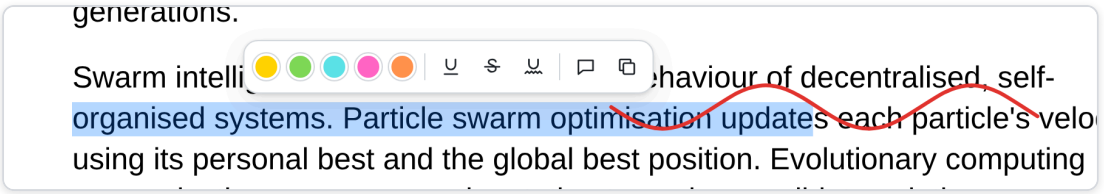
Quick popup on selected text

In Select mode, select text with the mouse (see *Selecting text*, section 6). A small bar appears above it:

- five colour dots — highlight in that colour;
- Underline, Strike-out, Squiggly;
- Comment — highlight and type a comment;

- Copy.

The popup can be switched off in Settings.



Quick popup after selecting text.

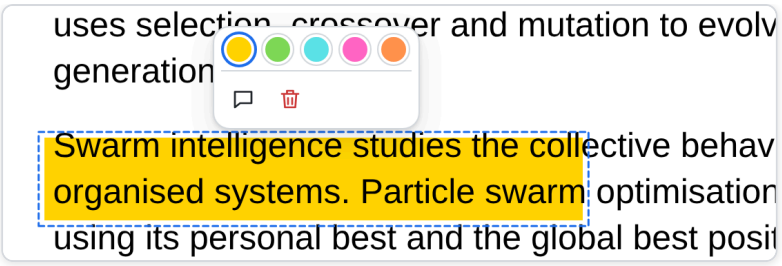
The tools

Tool	Key	How to use
Select	V	Select text; click an annotation to edit it; drag it to move it
Hand	H	Drag to move around the page
Highlight	1	Drag across text. On scanned pages, drag a box
Underline	2	Drag across text
Strike-out	3	Drag across text
Squiggly	4	Drag across text (wavy underline)
Pen	5	Draw freehand
Marker	6	Freehand see-through highlighter
Area highlight	7	Drag a rectangle anywhere — text, figures, equations, scanned pages
Eraser	E	Sweep over annotations to delete them (whole strokes or shapes)
Text	T	Click and type a text box (drag first to set its width); Esc or click outside to finish
Note	N	Click to place a sticky note and type its comment
Rectangle, Ellipse	—	Drag; hold Shift for a square or circle
Line, Arrow	—	Drag; hold Shift for 45° steps

Changing an annotation

In Select mode, click an annotation. A dashed box marks it and a small bar offers:

- colour dots;
- for highlights, underlines and strike-outs: switch between the four types;
- **Size** slider for pen, marker, shapes and lines;
- for text boxes: Edit text, smaller / larger font (or double-click the box to edit);
- Comment (add or edit a comment);
- Copy text (for highlighted text);
- Delete (or press Delete).



A selected annotation with its edit bar.

Drag a selected annotation to move it. A small coloured speech mark shows that an annotation has a comment. Right-click a page → **Add sticky note here** places a note exactly there.

Undo and redo

Ctrl+Z undoes and Ctrl+Y redoes any annotation change, move, delete or page rotation.

Compatibility

Annotations are saved as standard PDF annotations, so they appear in Adobe Acrobat, Chrome, Edge, Xodo and other readers. Annotations made in other readers can be edited in Ferna too.

8. Writing with a stylus (Xodo mode)

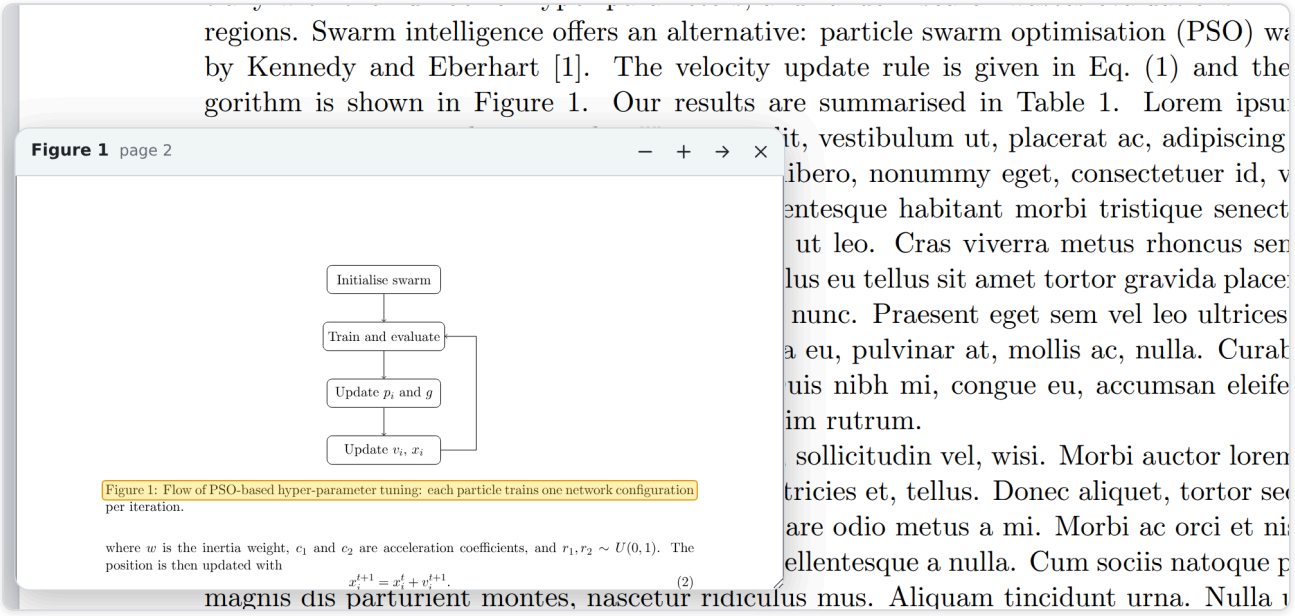
With the **Stylus** button on (it is on by default), a pen writes on the page at any time — you do not need to choose the Pen tool. The mouse and your finger keep scrolling and selecting text as usual.

- **Colour and thickness:** the stylus uses the Pen tool's settings. Open the panel while in Select mode; it shows them as **Stylus**.
- **Erasing:** turn the pen over and use its eraser end, or hold the eraser button, and sweep over strokes.
- **With another tool chosen:** the stylus uses that tool. For example, with Highlight chosen the stylus highlights text, with Rectangle it draws rectangles.
- **Palm rejection:** touches from your hand are ignored while the pen is on or near the screen.
- **Smooth ink:** every pen movement the screen reports is used, drawn on a fast low-delay layer with smoothed curves; strokes are saved as smooth curves too.
- **Turning it off:** click the Stylus button (or use Settings) if you want the pen to act like a mouse.

Ink has the same thickness along the whole stroke; pen pressure is not used.

9. Reference popups

Clicking a cross-reference shows the cited content in a floating card on top of the page you are reading, so you never lose your place.



Clicking “Figure 1” in the text shows the figure and its caption in a card.

- **Links:** click a figure, table, equation, citation or section link. The card shows the target page at that spot — a figure appears above its caption, the caption line is marked in yellow.
- **Plain text references:** in PDFs without links, click the words "Figure 3", "Fig. 2", "Table 1", "Eq. (4)", "Algorithm 1" or "Section 2.1". Ferna finds the caption or equation number and shows it.
- **In the card:** − / + make the preview smaller or larger, → goes to that page, × (or Esc, or a click elsewhere) closes it. Drag the card by its title bar; drag its bottom-right corner to resize it; scroll inside it to see more of the page.
- **Jumping instead:** Ctrl+click a link to go straight to the target. After any jump, a **Back** button appears at the bottom-left (or press Alt+←) to return to where you were.
- Switch the cards off in Settings → **Show cross-references as popups**.

10. Auto-Bookmark

Auto-Bookmark creates a bookmark tree (outline) for documents that have none, or replaces an existing one. It works for English and Sinhala documents and builds the tree the way a person would.

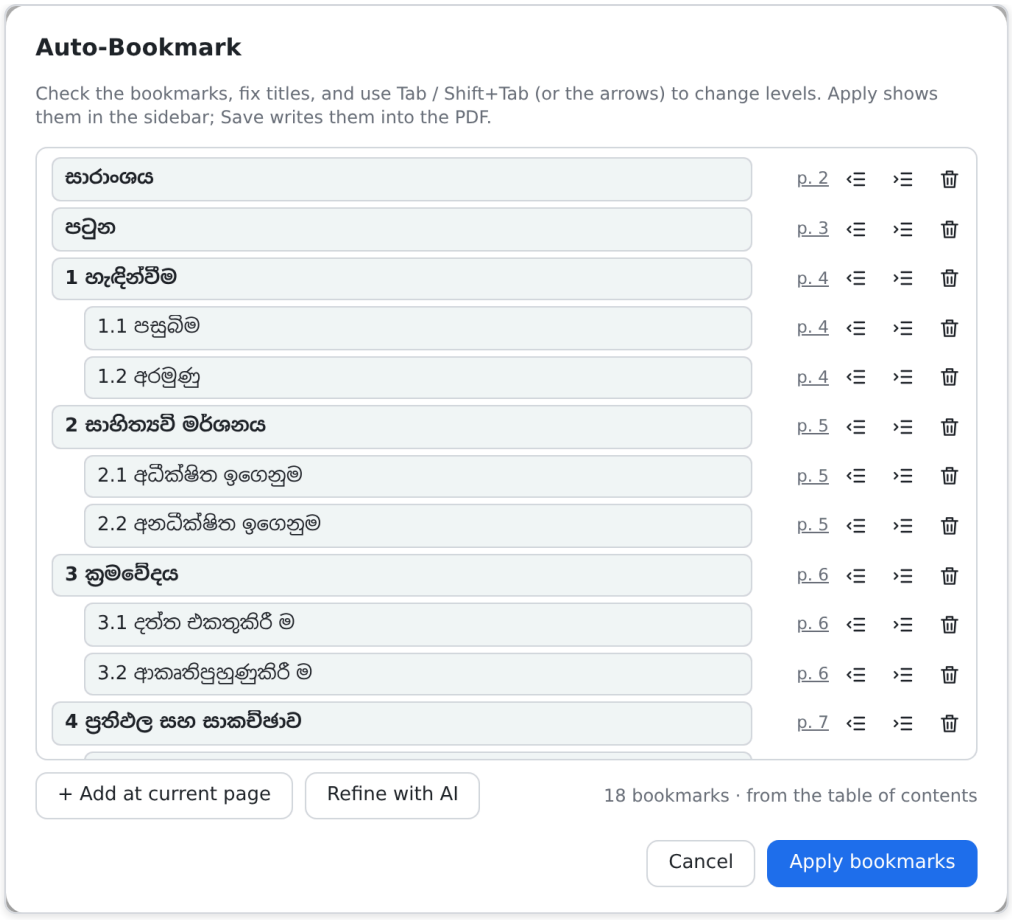
1. Click the **Auto-Bookmark** button in the toolbar (bookmark with a star), or Sidebar → Outline → **Auto-Bookmark**, or ⋮ → Auto-Bookmark.
2. **Table of contents first.** If the document has a printed table of contents (Contents, පටුන), Ferna reads it: the titles exactly as printed, their levels (from the numbering and indentation) and their page numbers. It works out how printed page numbers

(including roman numbers such as iii) relate to the real pages, and finds each heading on its page so that the bookmark lands exactly on it. The contents page itself, and front or back matter the contents list leaves out (Preface, Acknowledgements, සාරාංශය, References, ආශ්‍රිත ග්‍රන්ථ), are added in their place.

3. **Otherwise, from the page layout.** Ferna finds the headings from their numbering (1, 1.2, 1.2.3, I., A.), chapter words (Chapter 3, Appendix A, පරිච්ඡේදය 3, තෙවන පරිච්ඡේදය), common section names (Abstract, Introduction, References, හැඳින්වීම, ක්‍රමවේදය, නිගමනය …), font size and spacing. It leaves out numbered lists, figure and table captions (Figure, Table, රූපය, වගුව), running headers, title pages and author blocks; "Chapter 1" and the chapter title on the next line become one bookmark ("Chapter 1: Introduction"); ALL-CAPS headings are shown in normal title case; levels follow the numbering, else the font sizes.
4. **Slide decks** get one bookmark per new slide title (repeated "(cont.)" slides are skipped); section slides become the top level.
5. If the AI assistant is set up (section 11), the language model reviews layout-based lists: it removes lines that are not headings, repairs broken titles and chooses the levels, keeping every title in its own language. For a table of contents this step is not needed; click **Refine with AI** if you want it anyway.
6. Review the result in the editor:
    - click a title to edit it;
    - **Tab** / **Shift+Tab** (or the arrow buttons) move a bookmark one level in or out;
    - the bin removes it; **+ Add at current page** adds one; **p. N** shows that page;
    - **Refine with AI** / **Redo without AI** runs the other method.
7. Click **Apply bookmarks**. They appear in Sidebar → Outline at once.
8. **Save** (Ctrl+S) writes them into the PDF as real bookmarks that every PDF reader shows.

The line at the bottom of the editor tells you where the bookmarks came from: *from the table of contents*, *from slide titles*, *from page layout* or *refined by AI*.

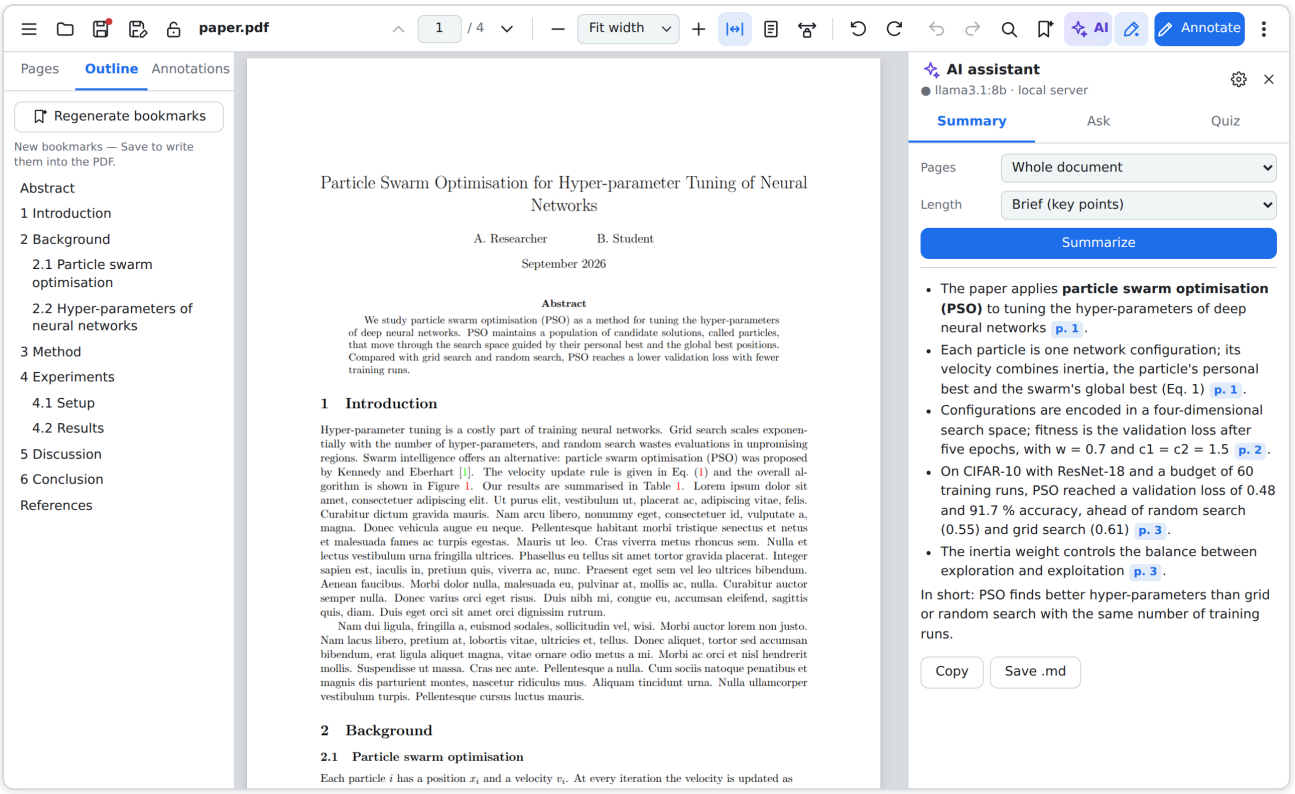
Slide decks get one bookmark per slide title.



Auto-Bookmark read from the table of contents of a Sinhala thesis.

11. AI assistant

Click ✦ **AI** in the toolbar (or press I) to open the AI panel on the right. It has three tabs.



AI panel with a brief summary (example output) and the generated bookmarks in the sidebar.

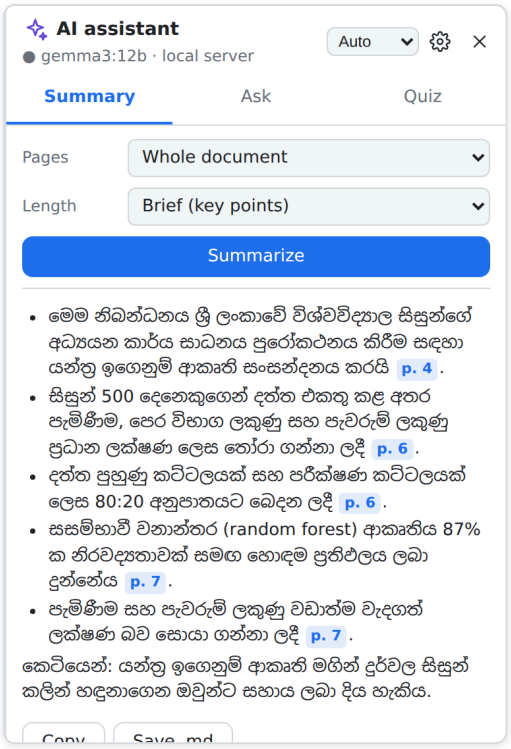
Tab	What it does
Summary	Summarises the whole document, the current page or a page range. **Brief** gives 5–7 key points; **Detailed** gives short sections. Copy it or save it as a Markdown (.md) file
Ask	A chat about the document. Answers are based only on the document and cite their pages — click **p. N** to go there. Quick buttons: Summarise this page, Key terms, Main contribution
Quiz	Generates 5–20 study questions: multiple choice (4 options with plausible wrong answers) and/or short answer, easy / moderate / hard, each with answer, explanation and page. Show answers one by one or all at once; copy or save as .md

Sinhala (සිංහල)

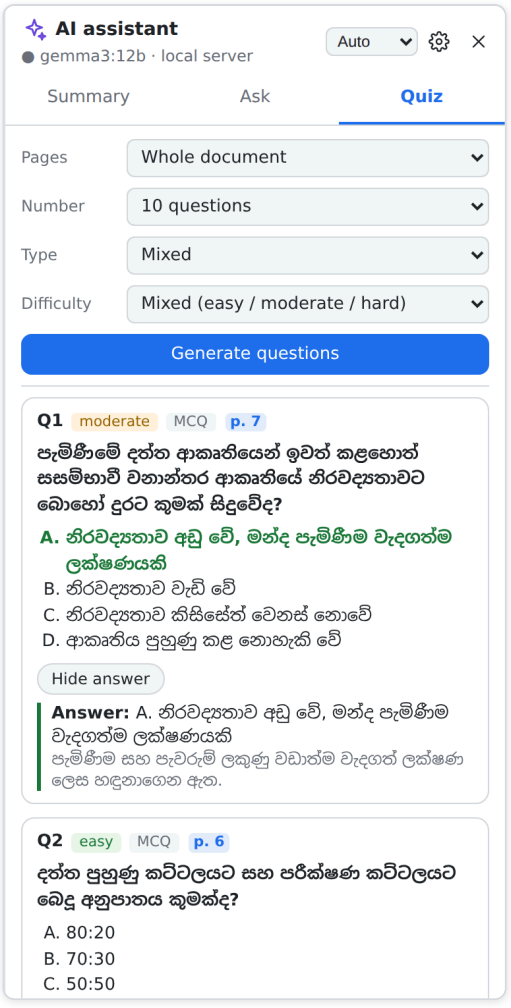
The AI works in Sinhala as well as English. The **language box** at the top of the AI panel sets the answer language:

Choice	Answers are written in
Auto (default)	Ask: the language of your question. Summary and Quiz: the language of the document
English	English, also for Sinhala documents
සිංහල	Sinhala, also for English documents

- You can ask in English about a Sinhala document (or the other way round): Ferna first turns your question into search words in the document's language, then answers.
- For Sinhala documents the quick buttons in Ask are in Sinhala, and **Ask AI** on a selected Sinhala passage asks for a Sinhala explanation.

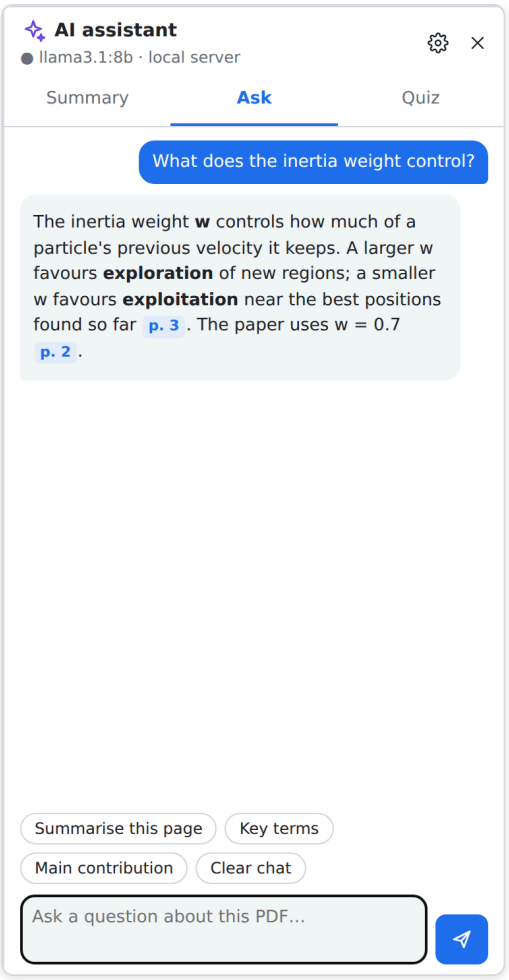


A Sinhala summary (example output).

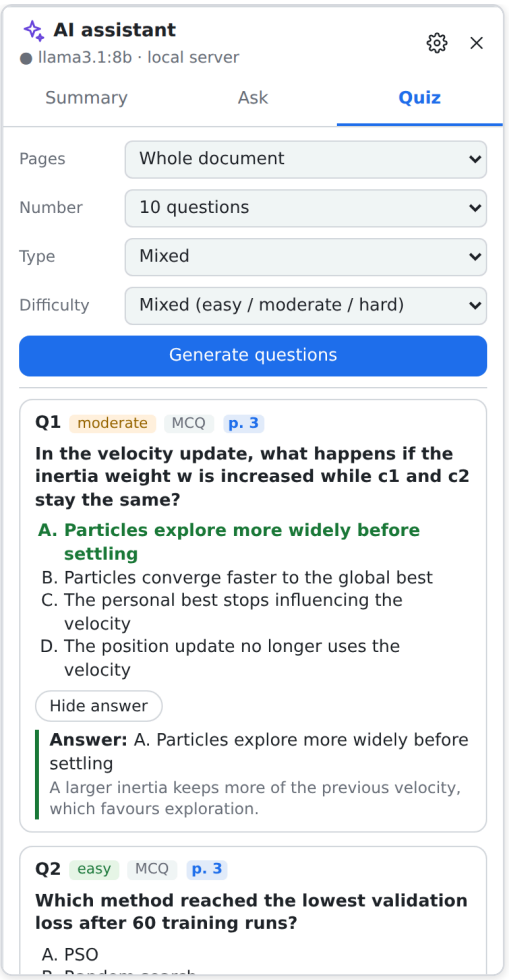


Sinhala study questions (example output).

- Sinhala text needs several times more of the model's working memory than English, so Ferna automatically reads smaller portions at a time and allows longer answers.
- **Choose a model that writes Sinhala.** On this device: **Qwen 3 · 4B** (best, needs about 3.4 GB of graphics memory) or **Qwen 3 · 1.7B** / **Gemma 2 · 2B** (smaller). The other on-device models mainly write English; the panel shows a yellow note if the chosen model is weak in Sinhala. With a local server, `gemma3:12b`, `gemma3:4b` or `qwen3:8b` give much better Sinhala. Chrome's built-in AI does not support Sinhala.



Ask: answers cite their pages.



Quiz: MCQ with answer and explanation.

Ask about a passage: select text on the page and click **Ask AI** in the quick popup; the assistant explains the passage in the context of the document.

Stop: while the AI is working, its button turns red and reads **Stop**.

First start: the on-device model

The AI runs on your own computer, on the graphics card (WebGPU). Nothing is sent to the internet.

1. The first time you use an AI feature, Ferna asks to download the model once. The default, **Qwen 2.5 · 1.5B**, is about 1 GB.
2. A progress bar under the panel title shows the download and start-up. Later starts take a few seconds and work offline.
3. The model stays loaded while the tab is open. **AI settings → Unload model** frees the graphics memory.

AI settings

Click the gear in the AI panel.

Option	When to use it
On this device (default)	Most computers. Choose the model: Qwen 2.5 1.5B (recommended for English, 1.0 GB), Qwen 3 1.7B (Sinhala + English, 1.1 GB), Qwen 3 4B (best Sinhala, 2.3 GB), Gemma 2 2B (multilingual, 1.4 GB), Llama 3.2 1B (fastest, 0.7 GB), Qwen 2.5 3B or Llama 3.2 3B (better English answers, 1.8 GB), Phi 3.5 mini (needs a strong graphics card). Downloaded models are listed and can be deleted
Chrome built-in AI	Google Chrome 138 or later with Gemini Nano; not available in Brave
Local AI server	Bigger models through Ollama or LM Studio on this computer. Ollama: install it from ollama.com, run `ollama pull llama3.1:8b` (for Sinhala: `ollama pull gemma3:12b`), keep the address `http://localhost:11434/v1` and enter the model name

Small on-device models can make mistakes. Check important answers against the cited pages. Scanned pages without a text layer cannot be read by the AI.

12. Saving, printing and passwords

Save — overwrites the file, no window

The PDF was opened…	Save (Ctrl+S) does this
From your disk (double-click, `file://` link)	Overwrites the original file directly, via the save helper (section 3)
With **Open** or by dragging it in	Overwrites the file directly (no helper needed)
From a website	First save: a copy in your **Downloads** folder (e.g. `paper.pdf`, or `paper (1).pdf` if that name exists). Later saves overwrite that copy

A red dot on the Save button means there are unsaved changes; closing the tab then asks for confirmation. If the save helper is not installed, Save on a disk PDF shows a short setup note and offers Save As. In Brave or Chrome, the setting "Ask where to save each file before downloading" makes website PDFs show a window again.

Save As

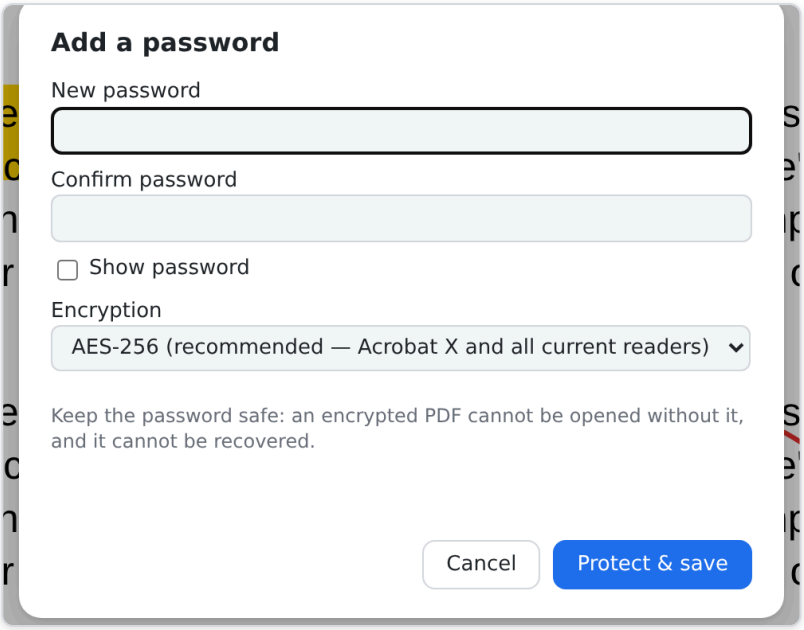
Save As (toolbar button, ⋮ menu or Ctrl+Shift+S) always asks for a file name and folder. The document then continues as that new file, so later Saves go to it.

Printing

Ctrl+P or ⋮ → **Print (with annotations)** prints the document including annotations and rotations. ⋮ → **Download original PDF** saves the file exactly as it was opened.

Passwords

Click the **lock** button (or ⋮ → Password…).



Adding a password.

1. **Add a password:** type it twice, choose the encryption and click **Protect & save**. The file is saved at once.
2. **Change or remove:** on a protected PDF the lock is highlighted; click it and choose **Change password…** or **Remove password**. The file is saved at once.

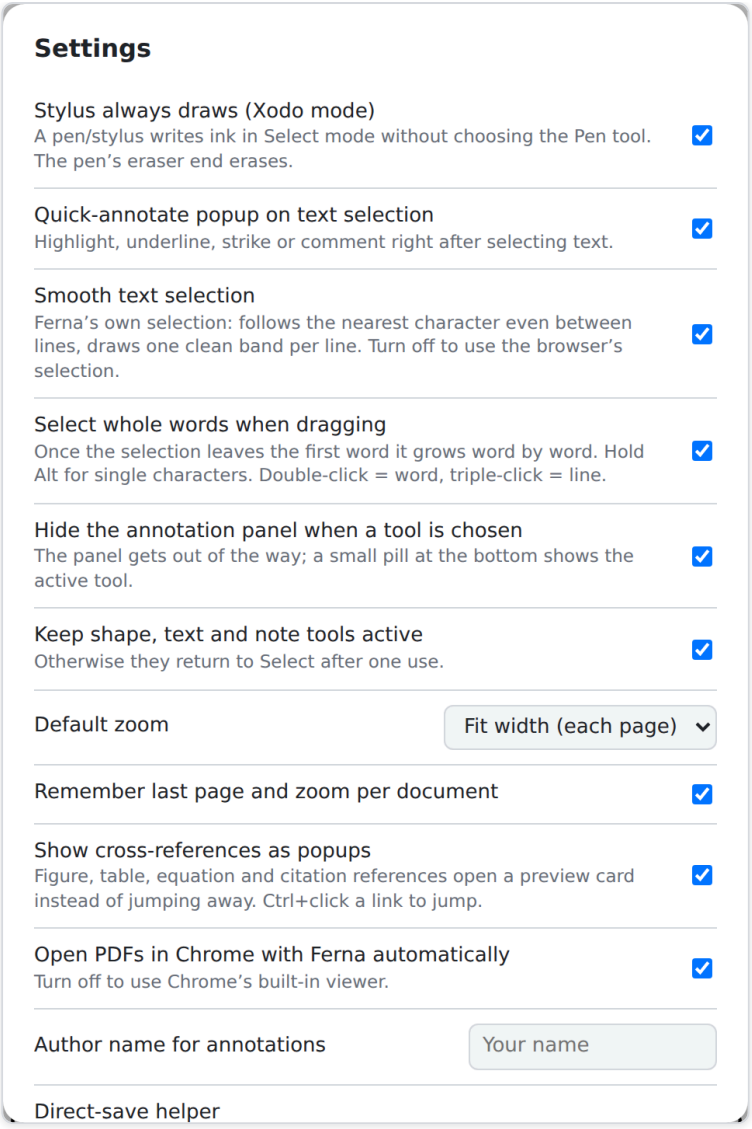
Encryption	Opens in
AES-256 (recommended, default)	Adobe Acrobat X and later, Chrome, Edge, Firefox, Xodo, Foxit, macOS Preview
AES-128	Also older readers (Acrobat 7 and later)

Opening a protected PDF asks for its password. You can then annotate it, and it stays protected with the same password when saved. Restrictions set by the document's author (for example, no printing) are kept.

> **A forgotten password cannot be recovered**, so keep it safe.

13. Settings

Open ⋮ **(More) → Settings**, change what you need and click **Save**.



The Settings window.

Setting	Default	Effect
Stylus always draws (Xodo mode)	On	A pen writes without choosing the Pen tool; its eraser end erases
Quick-annotate popup on text selection	On	Shows the highlight/underline/comment bar after selecting text
Hide the annotation panel when a tool is chosen	On	Panel closes on tool choice; the bottom bar shows the tool
Keep shape, text and note tools active	On	Off: these tools return to Select after one use
Default zoom	Fit width	Zoom for newly opened documents
Remember last page and zoom per document	On	Reopens each PDF where you left it

Setting	Default	Effect
Open PDFs automatically in Ferna	On	Off: the browser's own viewer opens PDFs (also via right-click on the toolbar icon)
Show cross-references as popups	On	Links and text references open a preview card; off: they jump to the target
Smooth text selection	On	Ferna's own selection with one band per line; off: the browser's selection
Select whole words when dragging	On	The selection grows word by word; hold Alt for single characters
Author name for annotations	Empty	Name stored with each annotation and shown in comments
Direct-save helper	—	Shows whether the save helper is installed

The Stylus and Horizontal lock buttons in the toolbar also switch their settings directly. The AI has its own settings (gear in the AI panel, section 11).

14. Keyboard shortcuts

Keys	Action
Ctrl+S	Save (overwrite)
Ctrl+Shift+S	Save As
Ctrl+O	Open a PDF
Ctrl+P	Print with annotations
Ctrl+F	Find
Ctrl+Z / Ctrl+Y	Undo / Redo
Ctrl + wheel, Ctrl + + / Ctrl + −	Zoom in / out
Ctrl+0 or W	Fit width
P	Fit page
Ctrl+1	100 %
L	Horizontal lock on/off
R / Shift+R	Rotate current page right / left
A	Show / hide annotation panel
V / H	Select / Hand
1 2 3 4	Highlight, Underline, Strike-out, Squiggly
5 / 6 / 7	Pen / Marker / Area highlight

Keys	Action
E	Eraser
T / N	Text box / Sticky note
Double-click / triple-click	Select a word / a line
Alt + drag	Select single characters instead of whole words
Delete	Delete the selected annotation
Esc	Close popup, then stop tool, then deselect
F4	Sidebar
I	AI assistant panel
Alt+←	Back after following a link
← / →	Previous / next page
?	Show the shortcut list

While typing in a text box, comment or field, these keys type normally.

15. Troubleshooting

Problem	Solution
PDF links open in the browser's own viewer	Check the extension is switched on; right-click the Ferna icon and tick **Open PDFs automatically in Ferna PDF Reader**
PDFs on my disk don't open in Ferna	On the extension's **Details** page switch on **Allow access to file URLs**
Brave downloads PDFs instead of opening them	`brave://settings/content/pdfDocuments` → **Open PDFs in Brave**
Save shows "One-time setup for direct saving"	Install the save helper (section 3), then restart the browser
Settings shows "Direct-save helper: Not installed" after installing	Restart the browser completely (close all windows). If it persists, run the installer again and read its message
Helper stopped working after updating from 1.0/1.1	Remove the extension and Load unpacked again (section 4), then reinstall the helper
"The file is read-only" when saving	Clear the file's Read-only attribute in File Explorer → Properties, or use Save As
Save fails while the PDF is open in another program	Close it in the other program, then save again

Problem	Solution
A website PDF saved to Downloads with "(1)" in the name	A file with that name already existed; later saves overwrite the " (1)" copy
A PDF from a site that needs a login won't open	Open it in the browser's own viewer (turn automatic opening off), download it, then open it with Ferna's Open button
Pen scrolls the page instead of writing	Check the Stylus button is on; hover the pen briefly before touching the screen
"This PDF's encryption cannot be rewritten"	The file uses an unusual protection scheme. You can read and annotate it but not save it; use Download original if needed
The AI asks to download a model	Normal on first use (about 1 GB, once). Choose a smaller model in AI settings if the connection is slow
"This browser has no WebGPU" / "no suitable graphics adapter"	Update the browser and graphics driver, or use AI settings → Local AI server (Ollama / LM Studio)
"The graphics card ran out of memory"	Choose a smaller model, close other AI tabs, or click Unload model
Ollama does not answer	Check it is running; set the environment variable `OLLAMA_ORIGINS=chrome-extension://*` and restart Ollama
Sinhala text copied from a PDF looks wrong in other programs, but fine in Ferna	Normal: Ferna repairs the text of such PDFs; other viewers show what is stored in the file
Sinhala text is shown as Latin letters (e.g. "wOHdmk")	The PDF uses an old non-Unicode Sinhala font (FM Abhaya and similar). Its text cannot be read as Sinhala; the page still displays correctly
AI answers in English although I asked for Sinhala, or the Sinhala is poor	Set the language box to සිංහල and choose Qwen 3 · 4B (or a local server model such as gemma3:12b) in AI settings
A Sinhala word has a space in the wrong place	Some PDF writers (for example XeLaTeX) store Sinhala word spaces slightly misplaced; Ferna repairs most cases but not all
A reference card shows the wrong place	The PDF's link points there; use → to jump or scroll inside the card
Annotations look different in another reader	Ferna saves standard annotations; small differences in fonts or line ends between readers are normal

16. Limitations, privacy and security

Limitations

- Direct saving of PDFs opened from disk needs the Windows save helper; on other systems use Save As.

- Ink has a uniform thickness; pen pressure is not used.
- Text boxes use Helvetica. Letters outside Western European alphabets (for example Sinhala) show correctly on screen but become "?" in the saved file.
- PDFs embedded inside web pages stay in the browser's own viewer; only PDFs opened as a full tab come to Ferna.
- Annotations already inside PDFs larger than 60 MB are displayed but cannot be edited.
- A forgotten PDF password cannot be recovered.
- Sinhala written with old non-Unicode fonts (FM Abhaya, DL-Manel and similar) cannot be read as text. In PDFs made with XeLaTeX some Sinhala word spaces may be misplaced.
- Small on-device models write limited Sinhala; larger models on a local server write it much better.
- The AI features need WebGPU and 1–4 GB of graphics memory (or a local AI server); small models can be wrong. Each open tab that uses the AI loads its own copy of the model.

Privacy and security

- Everything happens on your computer. Ferna sends no document, annotation or usage data anywhere. The AI model runs locally; the internet is used only to download it once (or to reach an AI server you configure yourself).
- Ferna fetches a PDF from its website only to display it, the same way the browser would.
- The save helper answers only this extension and only overwrites existing `.pdf` files.
- Ferna refuses to open documents inside other websites' frames.

Ferna PDF Reader is built on Mozilla pdf.js (Apache 2.0 licence), pdf-lib (MIT licence) and WebLLM (Apache 2.0 licence); their licence files are in the `lib` folder.